United Kingdom's Carbon Monoxide emissions Project

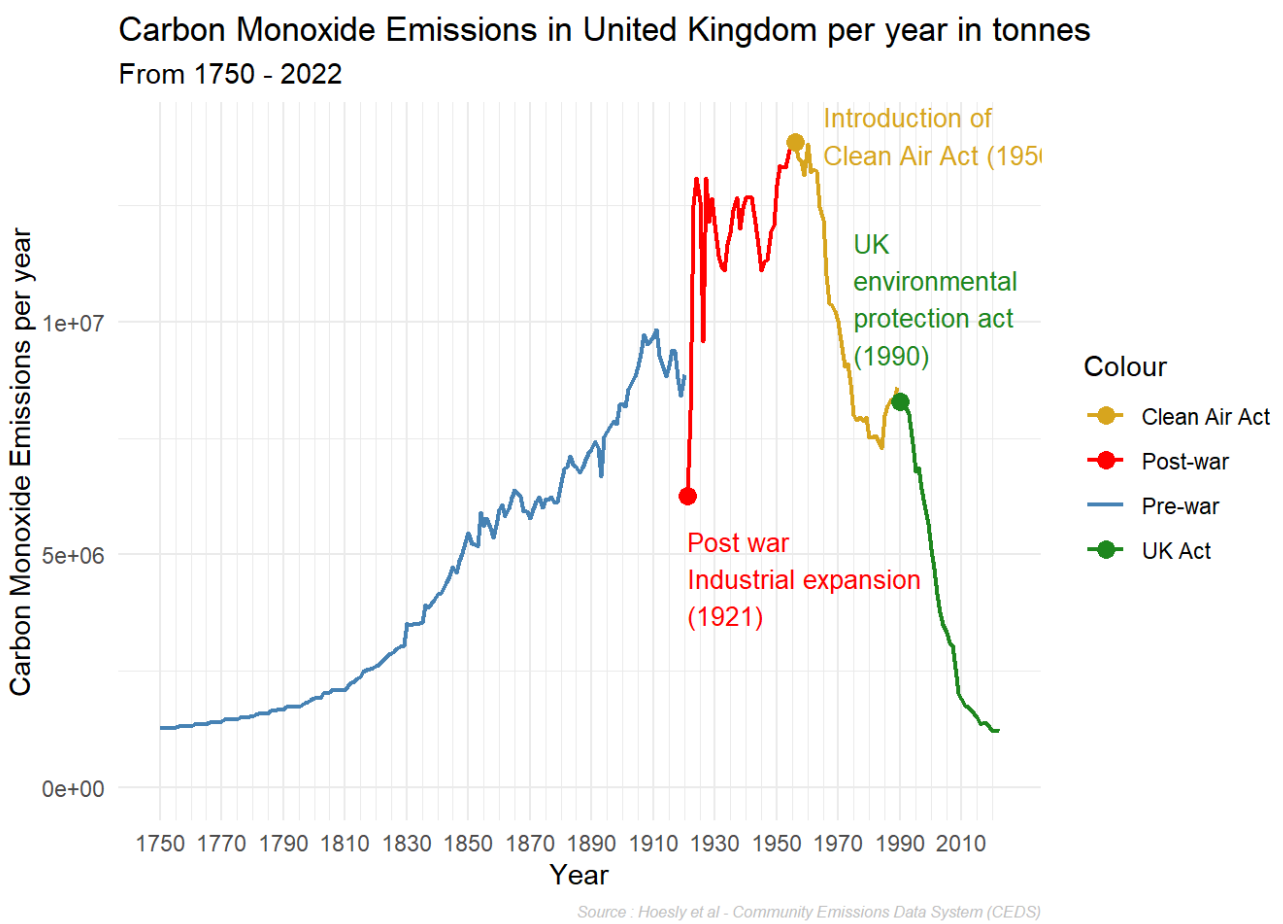
Question :- "What trends and turning points can be identified in United Kingdom's Carbon Monoxide emissions over time?"

```
# A tibble: 6 × 9
  entity      code   year emissions__pollutant_nox__sec…¹ emissions__pollutant…²
  <chr>       <chr> <dbl>                           <dbl>                  <dbl>
1 Afghanistan AFG    1750                            555.                   175.
2 Afghanistan AFG    1751                            558.                   176.
3 Afghanistan AFG    1752                            560.                   176.
4 Afghanistan AFG    1753                            562.                   177.
5 Afghanistan AFG    1754                            565.                   178.
6 Afghanistan AFG    1755                            567.                   178.
# ℹ abbreviated names: ¹emissions__pollutant_nox__sector_all_sectors,
#   ²emissions__pollutant_so2__sector_all_sectors
# ℹ 4 more variables: emissions__pollutant_co__sector_all_sectors <dbl>,
#   emissions__pollutant_bc__sector_all_sectors <dbl>,
#   emissions__pollutant_nh3__sector_all_sectors <dbl>,
#   emissions__pollutant_nmvoc__sector_all_sectors <dbl>
```

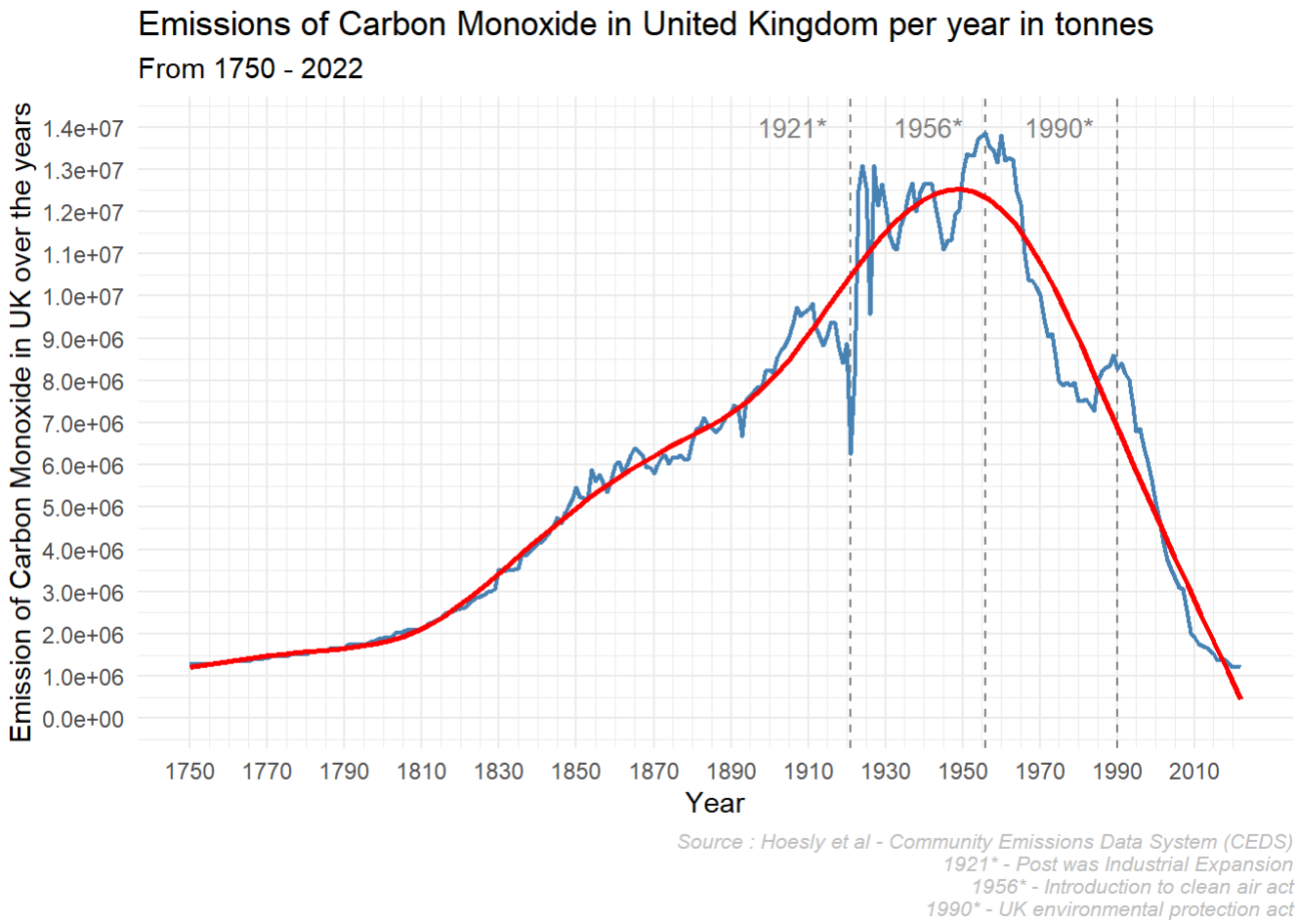
- Dataset now has data about various countries and regions.

```
entity   code   year     CO
     0      0      0      0
```

- Dataset now is filtered for United Kingdom and free of duplicates.



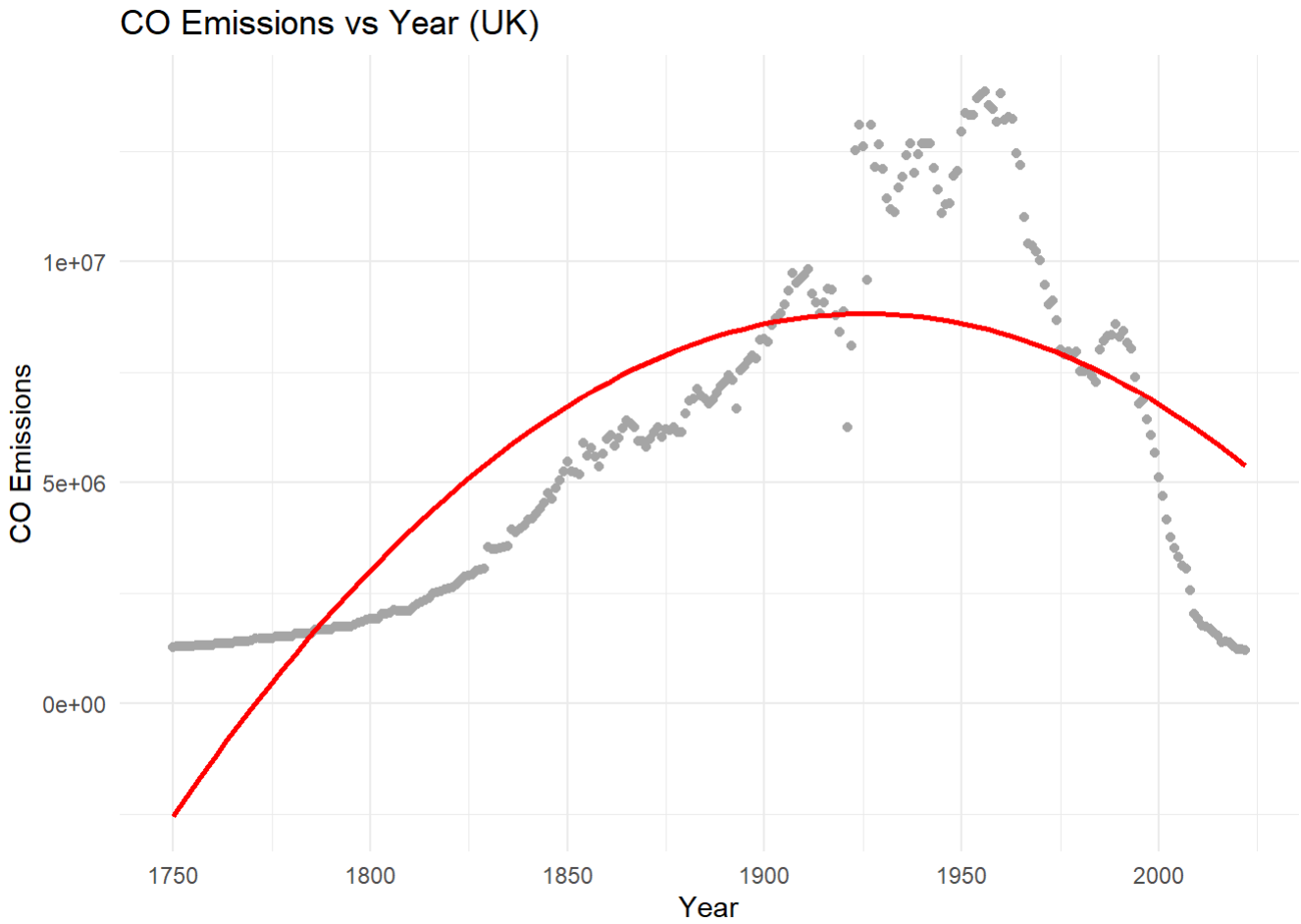
- Graph shows carbon monoxide emissions from 1750 to 2023. It clearly shows there was a serious increase in emissions and it reached aa peak at some year and then it started to reduce.
- Further, the graph shows different region of time where emissions has significant change. Red section are shows how after 1921's post-war industrial expansion emission started to increase rapidly.
- And it reached to its peak until Clean Air Act was introduced in 1956 and emissions decreased after that. Which is highlighted by the yellow region.
- Then, 1990's United Kingdom Environmental Protection Act was the final nail, from where it showed a further decrease. Which is highlighted by the green region.



- Smoothened graph clearly shows that the period between 1921 and 1956 had the most emissions. And after Clean Air Act and United Kingdom Environmental Protection Act there was a drastic decrease in emissions.
- It alos suggests that emssions trend is a curve and not a line.

```
Call:
lm(formula = CO ~ year + I(year^2), data = air_pollutants_UKCO)

Residuals:
     Min       1Q   Median       3Q      Max 
-4417534 -1727047  -413047  1441006  5421384 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.362e+09  9.180e+07  -14.84   <2e-16 ***
```

```
year         1.424e+06  9.747e+04   14.61   <2e-16 ***
I(year^2)   -3.698e+02  2.584e+01  -14.31   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2371000 on 270 degrees of freedom
Multiple R-squared:  0.6302,    Adjusted R-squared:  0.6275
F-statistic: 230.1 on 2 and 270 DF,  p-value: < 2.2e-16

    year
1925.342
```



- The positive coefficient for year and negative coefficient for year squared indicate an there was an increase at start and then followed by a decline after reaching a peak.
- The model explains approximately 63% of the variation in CO emissions (R^2 = 0.63), indicating a reasonably strong relationship between time and emissions.
- Model indicated year 1925.342 was the year of maximum emissions. This was the turning point from where emissions started to decrease.

Conclusion

- This analysis examined long-run carbon monoxide emissions in the United Kingdom.
- The visualisations show that emissions increased over time before reaching a peak and then declining in later years.
- A quadratic regression model supported this non-linear pattern, indicating that CO emissions did not follow a simple linear trend.

- Overall, the results suggest a strong historical shift in UK emissions over time, consistent with industrial growth followed by later environmental control and technological change.